

M4 Statement of Work

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Long-Term Objectives

- I. Enable stakeholders to replicate and further advance the payload deployment mechanisms to support future research missions involving ascent, descent, and ground-based payload releases.
- II. Reduce launch costs by incorporating multiple payload deployments into a single flight, thereby improving mission efficiency and resource utilization.
- III. Provide reliable and accurate flight data systems to monitor the rocket performance and individual payloads that can be implemented on future missions.
- IV. Support future development of specialized payloads that leverage these validated deployment mechanisms and data-collection capabilities during this current project.

Current Objectives

- I. Deliver a fully designed, high-power rocket with fully functioning avionics and payload deployment mechanism systems for payloads on descent and one ground payload, supported by electronic schematics, PCB layouts, and CAD drawings.
- II. Create and verify the flight software and telemetry system that implements data acquisition and a finite state machine capable of executing sequential payload deployments based on set altitude and time thresholds.
- III. Complete all necessary engineering documentation to support design decisions and project progression by delivering analysis and validation results for M5, M6 and final reports.

Technical Approach

To make a digital model of the rocket, the team will be using Solidworks. This lets us simulate the assembly and ensure the components fit together, which is crucial for a working mechanism. Some of the components will end up being 3D printed and, as such, can be taken directly from our initial designs. The fins and body tube will not be 3D printed; they will be fabricated out of plywood for the fins and kraft paper for the body tube. These materials are easy to work with, and a laser cutter will be used for the fins to ensure each one of the four required are identical. To design the PCBs, the team will be using KiCad. The electronics will be fabricated out of custom-designed PCBs, which will be ordered, and then the necessary components will be soldered onto them. Each electrical subsystem will be attached to a “raft,” which can be slid in and out of the rocket and positioned as needed in the electronics bay. The rest of the rocket will be assembled from purchased parts, such as the launch rail buttons, motor assembly, and ejection charges. Finally, the ground payload will be put into a passive gyroscopic ring assembly that can spin while the rocket is lying on its side to ensure that the payload is ejected in the proper orientation.

Project Plan

Start Date	End Date	Task Name	Task Description
1/12	1/19	M4	Create a statement of work for the semester.
1/12	1/31	Component Modeling	Draft the first CAD models of the project to prepare for component and system testing.
2/1	2/21	Electronics Testing	The ECE students will manufacture and test their custom PCBs to verify designs.
2/1	2/21	In-flight Deployment Prototyping	The concept for the in-flight deployment mechanism will be tested through rapid prototyping and laboratory experiments to verify the design efficacy.
2/1	2/21	Ground Deployment Prototyping	The concept for the ground deployment mechanism will be tested through rapid prototyping and laboratory experiments to verify the design efficacy.
2/1	2/21	Recovery System Testing	The chosen parachutes and their means of deployment must be tested in a laboratory setting to ensure they can slow the rocket's descent to the desired speed.
1/19	2/23	FMEA	The team's failure modes and effects analysis must be updated throughout the design process to identify the areas of concern where the project might encounter problems. These failure modes must be tested in the lab to ensure the design properly addresses all issues.
2/8	2/23	M5	Critical Design Evaluation
2/24	3/5	Integration Testing	With individual components of the rocket's design verified, the team must test to ensure all components work together as intended.
3/1	3/13	Manufacturing and Assembly	Complete the manufacturing process for fully tested components in preparation for the March 14th launch date.
3/14	3/14	Primary Launch Window	The team aims for the March 14th launch window at Palm Bay. All testing and manufacturing must be completed before this date. The launch will serve as the final prototype test to ensure the design functions through all stages.
4/11	4/11	Secondary Launch Window	If the March 14th date cannot be achieved, this launch window serves as a backup opportunity.
3/14	4/17	M6	Work will begin on the Milestone 6 report as soon as the launch is complete.